Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

Automated neonatal hip ultrasound system for diagnosing developmental dysplasia of hips using assistive AI

Young Seop Lee

liive2925@gmail.com

Gachon University

Young Jae Kim

Gachon University Gil Hospital

Jeong Won Ryu

Advanced Medical Technology Laboratory, Healcerion Co., Ltd.

Kwang Gi Kim

Gachon University Gil Hospital

Article

Keywords:

Posted Date: August 16th, 2024

DOI: https://doi.org/10.21203/rs.3.rs-4753646/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

Automated neonatal hip ultrasound system for diagnosing developmental dysplasia of hips using assistive AI

Young Seop Lee[1,*], **Young Jae Kim**[1,+], **Jeong Won Ryu**[2,+], **and Kwang Gi Kim**[1,3,-]

[1]Gachon Biomedical & Convergence Institute, Gachon University Gil Medical Center, Incheon, South of Korea
[2]Advanced Medical Technology Laboratory, Healcerion Co., Ltd., Seoul, South of Korea
[3]Department of Biomedical Engineering, College of Medicine, Gachon University, Incheon, South of Korea
[*]liive2925@gmail.com
[-]kimkg@gachon.ac.kr
[+]these authors contributed equally to this work

ABSTRACT

Children with developmental dysplasia of the hip (DDH) joint often do not exhibit clinical symptoms, and diagnosis in some cases relies solely on hip joint examination. There are several methods to examine the hip joint, with ultrasound being the simplest and most convenient diagnostic method. The Graff algorithm is commonly used to diagnose hip dysplasia by measuring angles at different landmarks in the hip region. This study utilized the SONON 300L device developed by Hilserion to acquire ultrasound images of the hip joints in 18 infants and children. Following classification, the standard planes were aligned with the Graf's landmarks using an artificial intelligence (AI) model, and the landmark regions were subsequently segmented. Significant findings were observed in the ultrasound images, with a maximum area-under-the-curve of 0.864 and a Dice coefficient of 0.660. The Graf algorithm was employed to develop an automated model for diagnosing DDH, resulting in a DDH-assisted AI model with an average Graf angle error rate of 0.21 compared to expert diagnostics

Introduction

Children with developmental dysplasia of the hip (DDH) often have no clinical symptoms, and in some cases, it can only be detected through hip examination[1,2]. Studies indicate that 55.2 – 66.7% of DDH cases diagnosed through clinical hip screening were inconspicuous in ultrasound-based hip examinations[3,4]. Failure to diagnose DDH in its early stages can limit treatment options[5–8]. Furthermore, research conducted in South Australia reported that diagnosis and treatment beyond 3 months increased the likelihood of poor prognosis[6] and presented the risk of irreversible damage to the cartilaginous acetabulum[9]. Therefore, early diagnosis is crucial for the successful treatment of DDH[10,11]. Ultrasound-based hip screening has been shown to significantly contribute to the early diagnosis of DDH[1,12]. Consequently, the use of ultrasound scanning for hip screening has increased[6,9]. Despite these dedicated efforts, the issue of late diagnoses persists in ultrasound-based DDH screening programs[9]. This persistence can be attributed to factors such as operator dependency, insufficient equipment training, or the occurrence of late-onset DDH rather than cases being "missed" during screening[9].

The Graf method of infantile hip ultrasonography[13,14] was the first standardized approach and is perhaps the most widely used[15]. This method involves the anatomical identification of the four key structures (in the respective order): the chondro-osseous junction, femoral head, synovial fold, and the joint capsule. Following this three landmarks are identified: the lower limb (os ilium), mid portion of the acetabular roof, and the labrum. When all three landmarks are visible, the hip sonogram is considered to be in the " standard" plane[14]. However, distinguishing these landmarks during standard planar selection is time-consuming and can lead to inaccurate and subjective results[16]. Angle measurements also suffer from high intra- and inter-observer variability[17].

Automating the identification process could yield considerable time-saving benefits. However, to the best of our knowledge, there has been limited research in this specific area . In 2019, Alan et al.[18] introduced a support vector machine classification model to classify four degrees of hip displacement—normal hip, subluxation, low dislocation, and high dislocation—achieving an overall accuracy of 79%. Notably, this study used computer- generated samples where a model of the hip joint was mechanically moved to generate data. Nevertheless, this study is significant, as it introduces the potential of machine learning applications in identifying hip displacement . In 2023, Jaremko et al.[19] proposed integrating an artificial intelligence (AI)-aided workflow into clinical settings. They introduced a U-Net based network[20] to segment three landmarks using infantile hip ultrasonography. The performance evaluation of the model was not presented. However, they reported a reduction in follow-up

scans due to “suboptimal” scans or AI-recommended rescans.

We present a novel approach to infant hip ultrasonography, supported by Hilserion’s ultrasound device integrated with a smartphone. Our goal is to implement an AI capable of diagnosing DDH utilizing smartphone resources effectively. Despite smartphones’ lower computational power compared to personal computers (PCs), we developed a DDH reading model optimized for maximum performance with minimal parameters. To build an automated diagnostic system for DDH, we designed and evaluated an AI model that can classify standard planes and perform semantic segmentation for Graf discrimination. Our study aims to streamline DDH diagnosis using image processing and Graf discrimination, addressing challenges related to time and expertise. Schematic diagram of automated neonatal hip ultrasound for diagnosis of developmental hip dysplasia using assisted AI process is shown in Figure 1.

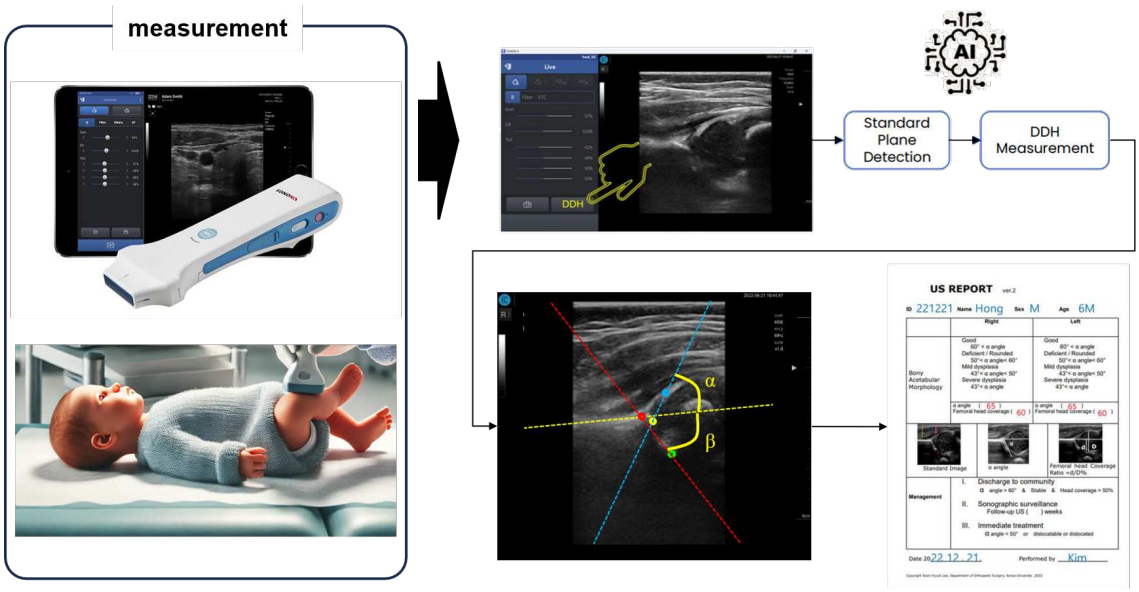


Figure 1. Schematic diagram of automated neonatal hip ultrasound for diagnosis of developmental hip dysplasia using assisted AI.

Results

The study involved three main tasks related to infant hip ultrasonography. Firstly, real-time classification of raw ultrasound images into standard planes was achieved using a convolutional neural network (CNN) model. To ensure compatibility with mobile devices and low-spec PCs, the CNN model was designed with limited parameters and computational demands. Secondly, image region segmentation was performed to facilitate application of the Graf algorithm. In this step, the regions of the image are segmented, forming the basis for applying the graph algorithm for hip dysplasia assessment. Thirdly, DDH was conducted using Graf algorithm on the segmented regions. For the standard plane classification task, 3476 test sets were used, including 584 standard planes. Figure 2 visually depicts the receiver operating characteristic (ROC) area under the curve (AUC) (ROC-AUC) for the standard plane classification. ROC-AUC is one of the most important indicators of a model’s classification ability. It helps us holistically understand the results of the study .

Figure 2 presents the ROC curves for several models, including DenseNet121, EfficientNetV2B0, MobileNetV1, MobileNetV2, ResNet50, and NASNetMobile. NASNetMobile demonstrated the best learning performance, achieving the highest AUC at 0.864. Standard Plane classification Model Performance in Table 1 further illustrates that compared to other models, NASNetMobile offers distinct advantages in terms of computational efficiency and parameter count. Furthermore, the comprehensive classification performance results reinforce the superiority of NASNetMobile over alternative models. Its AUC of 0.864 indicates robust capability in accurately classifying and distinguishing the dataset. These findings in Standard Plane classification Model Performance in Table 1 complement the ROC plot in Figure 2, providing a detailed assessment of the models’ relative performance.

Standard Plane classification Model Performance in Table 1 presents the performance of the standard plane classification model, accompanied with 95% confidence intervals (CIs). Notably, CI primarily serves as a descriptive statistical measure but also facilitates inferences about the broader population. The 95% CI values for each performance metric were consistently below 0.04, indicating statistical stability and minimal variation in the model’s performance estimates. A narrower CI indicates

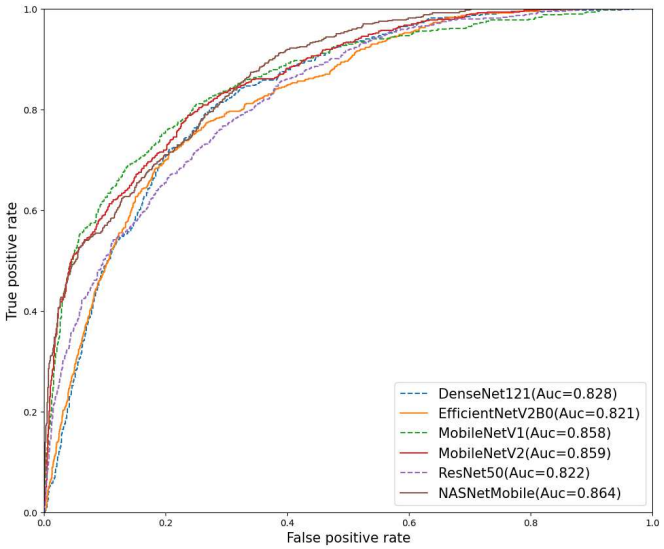


Figure 2. Standard flat classification AI model ROC graph.

Table 1. Standard Plane classification and Graf Region Segmentation Model Performance with 95% CIs

Standard Plane classification Model Performance				
Model	**AUC**	**Sensitivity**	**Specificity**	**F1-Score**
MobileNetV2	0.858 (0.843 – 0.874)	0.880 (0.869 – 0.891)	0.973 (0.967 – 0.978)	0.880 (0.869 – 0.891)
MobileNetV1	0.857 (0.840 – 0.874)	0.880 (0.869 – 0.890)	0.960 (0.952 – 0.967)	0.880 (0.869 – 0.890)
DenseNet121	0.827 (0.811 – 0.844)	0.835 (0.822 – 0.848)	0.950 (0.942 – 0.958)	0.835 (0.822 – 0.847)
EfficientNetV2B0	0.822 (0.804 – 0.839)	0.839 (0.827 – 0.851)	0.943 (0.934 – 0.951)	0.839 (0.828 – 0.851)
NASNetMobile	0.864 (0.850 – 0.878)	0.880 (0.869 – 0.891)	0.974 (0.968 – 0.980)	0.881 (0.869 – 0.892)
ResNet50	0.821 (0.804 – 0.839)	0.857 (0.845 – 0.868)	0.969 (0.963 – 0.976)	0.857 (0.847 – 0.868)
Graf Region Segmentation Model Performance				
Model	**Dice-Coefficient**	**Sensitivity**	**Precision**	**Specificity**
NestedUNet	0.629 (0.590 – 0.668)	0.583 (0.543 – 0.623)	0.688 (0.651 – 0.726)	0.998 (0.995 – 1.002)
UNet	0.628 (0.589 – 0.667)	0.572 (0.531 – 0.612)	0.699 (0.661 – 0.736)	0.998 (0.995 – 1.002)
DeepLabV3Plus	0.660 (0.622 – 0.699)	0.621 (0.582 – 0.660)	0.705 (0.668 – 0.742)	0.998 (0.995 – 1.002)
PAN	0.612 (0.572 – 0.651)	0.567 (0.527 – 0.607)	0.669 (0.630 – 0.707)	0.998 (0.995 – 1.002)

a more robust and accurate model[21]. Given the imbalanced nature of the test data in this study, the F1-Score was selected as the performance metric of interest. Remarkably, both MobileNetV2 and NASNetMobile demonstrated superior performance. The 95% CI of the F1-Score for each model was 0.011, providing a comprehensive assessment that encompassed both precision and recall. This statistical measure offers valuable insights into the reliability and generalizability of a model. The CI of the F1-Score proves particularly useful in identifying and comparing performance disparities between models. This analysis underscores the stability and accuracy of the selected plane classification model. It also highlights the effectiveness of MobileNetV2 and NASNetMobile, which outperformed other models in terms of F1-Score within the established CI. Including CIs enhances the statistical robustness of the assessment, offering a more nuanced understanding of the relative performance of the different models.

In the context of future applications involving devices and programs, the utilization of the MobileNetV2 model is envisaged. Specifically, we will focus on executing real-time standard plane classification when deployed on devices. Our classification methodology will incorporate a unique approach: if a specific input is consistently identified as a standard plane across five consecutive frames, the subsequent prediction will transition to the Graf area division model for the subsequent five frames. Graf Region Segmentation Model Performance in Table 1 showcases the performance metrics associated with the Graf area division model, providing a comprehensive overview of its efficacy in handling diverse scenarios and classification tasks. This

strategic integration of models aims to enhance the overall efficiency and system accuracy, thereby ensuring a robust and adaptive response to dynamic inputs over time.

Graf Region Segmentation Model Performance in Table 1 provides a detailed representation that encapsulates both the performance metrics and the 95% CI for the Graf region segmentation model. This model incorporates a sophisticated ensemble of architectural frameworks, including NestedUNet, UNet, DeepLabV3Plus and PAN. These architectural choices represent a deliberate and comprehensive approach to leveraging different deep learning techniques for robust image segmentation. The evaluation of model performance involves the application of the Dice coefficient technique, which assesses the degree of overlap and statistical significance between pairs of reference and candidate image objects[22]. Notably, within this ensemble, DeepLabV3Plus emerged as the best performer, achieving the highest Dice coefficient. The associated CI interval, measured at 0.038, underscores the statistical significance of this performance estimate. The interpretation of this CI interval is crucial. Its narrow width suggests that the performance of the DeepLabV3Plus model exhibits minimal variation, indicating a high degree of stability. Statistically, a narrower CI indicates a robust and consistently accurate model. This stability is critical for real-world applications, ensuring reliable and predictable performance despite potential variations in input data. In essence, these results highlight the effectiveness of DeepLabV3Plus in achieving both statistical significance and operational stability in Graf region segmentation. These findings position the Graf Region segmentation model, particularly that leveraging DeepLabV3Plus, as a promising solution for image segmentation tasks, laying the groundwork for further advancements in the field.

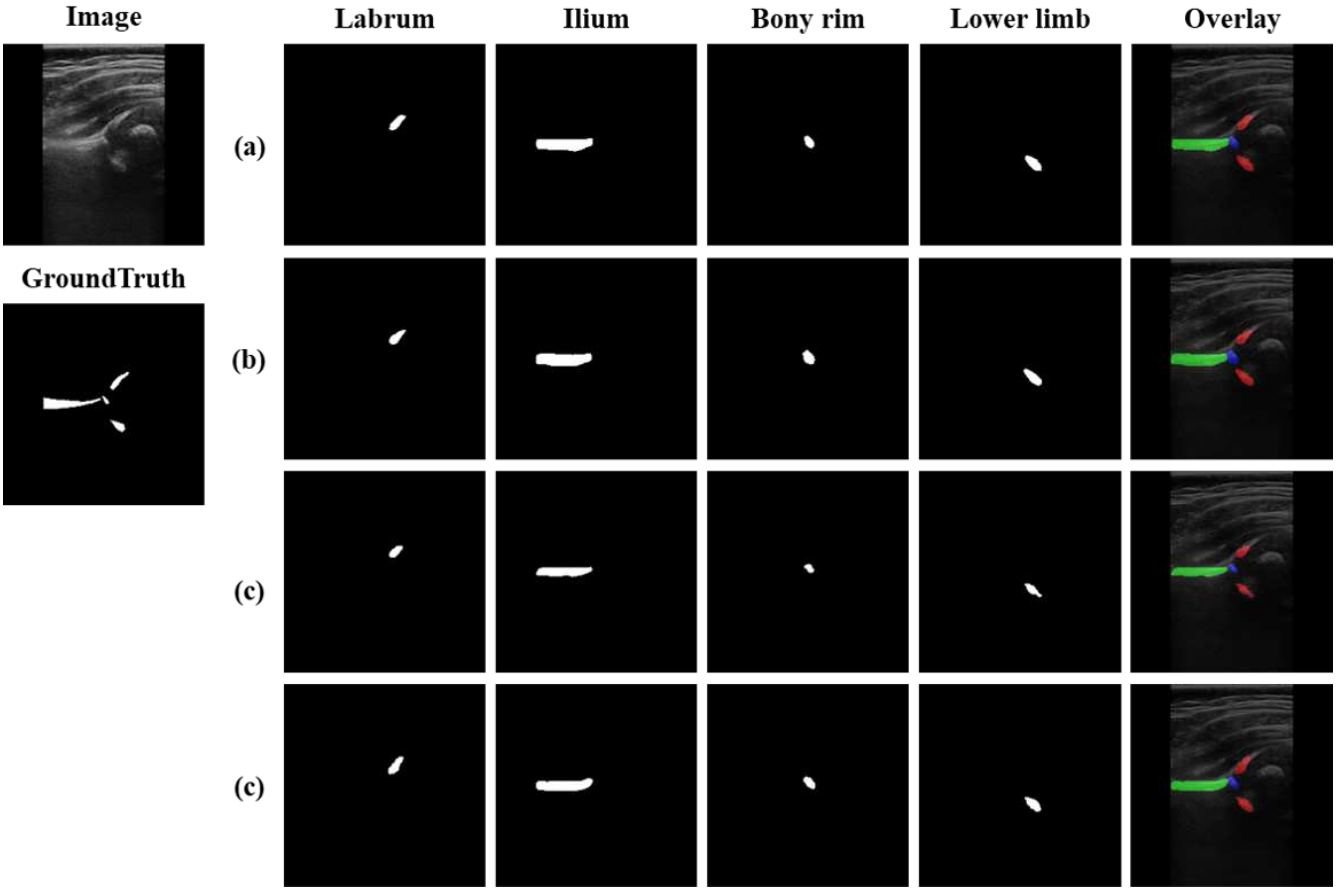


Figure 3. Graf Region Segmentation Model Results (a) NestedUNet Predict result (b) UNet Predict result (c) DeepLabV3Plus Predict result (d) PAN Predict result

Figure 3 visually depicts the outcomes of Graf region segmentation analysis, focusing on masks obtained for the labrum and ilium, bony rim, and lower limb. The validity of all results was confirmed through a comprehensive assessment of accurate localization, demonstrated by the overlay of predictions. Specifically, Figure 3(c) showcases the model's prediction mask generated by DeepLabV3Plus for the lower limb, revealing an omission at the borderline portion of the femoral head. This observation suggests potential for accurate results in future applications using the Graf algorithm. The visualization in Figure 3 serves as a valuable diagnostic tool that guides the refinement of the segmentation process. It highlights the model's capability to identify specific anatomical regions and illustrates the areas that may require improvement. The iterative improvement

process, driven by visual analyses similar to those presented in Figure 3, contributes to continuous enhancements in the model's accuracy and efficiency for Graf region segmentation tasks. In summary, the results presented in Figure 3 underscore both the strengths of the model and areas for improvement, laying the foundation for advancing accuracy and effectiveness in future developments.

The primary objective of this study is the practical application of the developed model using SONON X devices, designed by Hilserion Inc. of South Korea. The planned implementations involve ultrasound scanning facilitated by SONON 300L handheld linear probes. The adaptability of the model extends to various hardware configurations, emphasizing compatibility with low-spec PCs and mobile devices. To evaluate the performance of the model in real-world conditions, extensive testing was conducted using the SONON X device application. Specifically, the model was evaluated based on ultrasound scans processed by SONON 300L handheld linear probes. Notably, a decision was made to deliberately exclude dedicated graphic card resources, performing tests solely using CPU resources. This approach was adopted to simulate real-world scenarios where low-spec PCs and mobile devices lack access to dedicated graphics processing units. The CPU resources utilized in the tests was an Intel(R) Core (TM) i9-10900 CPU @ 2.80GHz. This configuration served as the computational backbone of the model, influencing critical parameters and processing speed. Table 2 provides a detailed analysis of these parameters and the computational speeds, offering insights into the efficiency and adaptability of the model across different hardware specifications.

Table 2. Model performance and device coverage performance

Device-Applied Classification Performance				
Model	**F1-Score**	**Total params**	**Run time(ms)**	**St. Dev.**
MobileNetV2	0.858 (0.843 – 0.874)	3.601*M*	6.718	0.979
MobileNetV1	0.857 (0.840 – 0.874)	4.324*M*	5.188	1.117
DenseNet121	0.827 (0.811 – 0.844)	8.127*M*	19.750	2.831
EfficientNetV2B0	0.822 (0.804 – 0.839)	7.262*M*	6.219	1.272
NASNetMobile	0.864 (0.850 – 0.878)	5.391*M*	6.187	1.093
ResNet50	0.821 (0.804 – 0.839)	25.697*M*	16.219	2.692
Device-Applied Segmentation Performance				
Model	**Dice-Coefficient**	**Total params**	**Run time(ms)**	**St. Dev.**
NestedUNet	0.629 (0.590 – 0.668)	6.825*M*	46.212	6.081
UNet	0.628 (0.589 – 0.667)	6.629*M*	33.096	5.434
DeepLabV3Plus	0.660 (0.622 – 0.699)	4.379*M*	40.078	7.099
PAN	0.612 (0.572 – 0.651)	2.471*M*	39.073	6.939

Table 2 provides a comprehensive overview of the model performance in the context of device applications. Notably, MobileNetV2 emerged as the top performer in the standard plane classification model, demonstrating its effectiveness in accurately categorizing standard planes. Despite MobileNetV1's lower runtime performance, MobileNetV2 demonstrated remarkable efficiency, completing computations in only 6.718 ms using CPU resources. This remarkable speed indicates the model's ability to perform real-time classification, highlighting its suitability for use in both PCs and mobile devices. This adaptability aligns with the goal of ensuring the model's usefulness across different hardware configurations, especially those characterized by resource constraints. Moreover, the analysis of runtime standard deviation (SD) provides additional insights into the model's stability. In particular, MobileNetV2 exhibited the lowest runtime SD, indicating a high degree of performance stability, with minimal variation in runtime over multiple executions. Such model consistency is critical in real-world applications where reliable and predictable performance is paramount.

A closer look at the performance of the Graf region segmentation models, as shown in Table 2, reveals valuable insights. DeepLabV3 Plus emerges as the top performer, with the highest Dice Coefficient among the evaluated models. This indicates its superior accuracy in segmenting Graf regions, demonstrating its effectiveness in capturing fine details and intricate structures within the images. However, a trade-off arise when considering the runtime performance of DeepLabV3 Plus. With a runtime of 40.078 ms and an SD of 7.099, it is evident that this model requires relatively more computational resources and exhibits some variability in execution times. For Graf region segmentation, the runtime may not be as significant as that of the model operating individually after image acquisition, rather than in real-time processing scenarios. This implies that the execution time of the model is not a critical factor in determining its usefulness for this segmentation task. In this context, the emphasis shifts to accuracy and precision in delineating Graf regions, where DeepLabV3 Plus excels. Therefore, for the Graf region

segmentation model, the choice of DeepLabV3 Plus seems justified. Despite its relatively longer runtime, the exceptional segmentation performance of the model, as well as the less critical nature of the runtime in this specific application, make it an appropriate choice. This nuanced evaluation highlights the importance of aligning the model selection with the unique requirements and constraints of the task at hand within the overall application framework.

Table 3. Results of using the Graf algorithm and AI for each patient

Performance	Test Set 1	Test Set 2	Test Set 1 & Test Set 2
Mean ($\pm$ std.) α	$62.34^\circ \pm 2.69^\circ$	$63.11^\circ \pm 2.45^\circ$	$62.74^\circ \pm 2.56^\circ$
AI Mean ($\pm$ std.) α	$62.85^\circ \pm 2.24^\circ$	$62.99^\circ \pm 2.14^\circ$	$62.92^\circ \pm 2.19^\circ$
Mean ($\pm$ std.) β	$53.22^\circ \pm 1.78^\circ$	$52.71^\circ \pm 2.00^\circ$	$52.98^\circ \pm 1.89^\circ$
AI Mean ($\pm$ std.) β	$53.28^\circ \pm 1.45^\circ$	$53.01^\circ \pm 1.17^\circ$	$53.15^\circ \pm 1.33^\circ$

Table 3 compares the results of using the Graf algorithm by medical professionals and AI for each patient. Both patients tested negative for DDH, with α tests being >60 and β tests being <55. Importantly, the application results of the Graf algorithm in the AI automation model showed an average difference of 0.248° compared to the measurements by the professional medical staff. Additionally, a lower SD was observed, which is statistically significant.

Discussion

The overall aim of this study is to apply the developed model to the SONON X device application, with a specific focus on ultrasound scans facilitated by the SONON 300L handheld linear probe. The intended application extends to low specification PCs and mobile devices, in line with the wider aim of ensuring accessibility and versatility in healthcare technology.

The overall aim of this study was to apply the developed model in the SONON X device application, specifically focusing on ultrasound scans facilitated by the SONON 300L handheld linear probe. The intended application extends to low-spec PCs and mobile devices, aligning with the wider objective of ensuring accessibility and versatility in healthcare technology. A significant challenge highlighted by this study concerns the limitations of data extracted using the SONON 300L device. While the dataset contains a substantial number of images, totaling 36,707, the small number of patients—only 18 cases—raises concerns about the model's reliability, particularly in avoiding the global minima during training. Global minima occur when the optimization algorithm converges to a suboptimal solution due to insufficient data diversity, potentially compromising the model's ability to generalize. Acknowledging this potential limitation, this study emphasizes the importance of clinical trial results and meticulous labeling efforts to mitigate concerns related to data scarcity. It is expected that rigorous clinical trials and accurate labeling of available data can significantly improve the reliability and generalizability of the model. In light of these considerations, this study underscores the need for future research to prioritize expanding the dataset to include a more diverse range of patient cases. This expansion could contribute to a more comprehensive understanding of model performance across different clinical scenarios. In addition, collaboration between healthcare professionals and domain experts is recommended to ensure that the model's outputs are consistent with clinical expectations and standards. A critical aspect highlighted in this study relates to the challenges associated with labeling the current dataset, which was identified as a factor affecting the performance of the Graf region segmentation model. The recognition of inconsistencies in labeling underlines the importance of addressing this issue to improve the overall effectiveness of the model. The suboptimal performance of the Graf region segmentation model, as revealed by the study, is directly related to inadequacies in current labeling practices. This finding underscores the need for corrective action to address labeling deficiencies. In addition, this study advocates a collaborative approach involving both data annotators and healthcare professionals to bridge the gap between technical requirements and clinical relevance. This collaborative effort can contribute to the development of a more refined and clinically meaningful dataset, fostering a stronger link between model predictions and real-world medical scenarios.

In conclusion, this study highlights the potential for improvement through conscientious data collection, robust clinical trials, and meticulous labeling practices despite the challenges posed by the limitations of the current dataset. Addressing these critical issues is imperative to strengthen the reliability and applicability of the models developed in the complex landscape of real-world healthcare, particularly within the dynamic framework of SONON X device applications. Furthermore, the correction of data labeling inconsistencies emerges as a paramount consideration to unlock the full potential of the Graf region segmentation model. The proposed solutions advocate a methodical relabeling process accompanied by an expert review, emphasizing the twin pillars of accuracy and clinical relevance. These nuanced corrective actions are essential to ensure that the model's performance consistently meets the stringent standards required for effective and reliable healthcare applications within a sophisticated SONON X device framework. In presenting these results, this study underscores the ongoing commitment to refine methodologies and advance technologies to improve patient care and diagnostic accuracy in medical imaging.

Methods

The data in this study were collected using SONON 300L from Healcerion Inc. The subjects were infants and toddlers less than one year of age, and data were obtained from ultrasound images of the hip joint. The Institutional Review Board (IRB) of Gachon University (IRB Number: GCIRB2020-477) approved this study and the requirement for informed consent was waived because of the retrospective nature of the study design. All experimental protocols were performed in accordance with the relevant guidelines and regulations in compliance with the Declaration of Helsinki. Images and labelling data supporting the findings of this study are available with permission from the corresponding author.

Table 4. DDH Ultrasound Dataset

Standard Plane Classification Dataset				
Class	**Train**	**Validation**	**Test**	**Total**
Patient Video Case	14	2	2	18
Standard Plane Frame	3,348	479	584	4,411
Not Standard Plane Frame	24,763	7,283	2,892	34,938
Aug. Standard Plane Frame	20,088	479	584	21,151
Aug. Not Standard Plane Frame	148,578	7,283	2,892	158,753
Graf Region Segmentation Dataset				
Class	**Train**	**Validation**	**Test**	**Total**
Patient Video Case	14	2	2	18
Graf Region mask Frame	3,348	479	584	4,411
Aug. Graf Region mask Frame	20,088	479	584	21,151

Table 4 lists the number of datasets constructed using the acquired data. The image were captured at 30 frames per second. The training, validation, and test sets were constructed from ultrasound images of 14, 2, and 2 patients, respectively. When the image was divided into frame units, an average of 2186 frames were extracted per image, with an average image duration of approximately 1 min 12 s. Training involved image augmentation across two environments, employing methods outlined in Figure 4 (a).

Figure 4(a) shows the image of pre-processing and augmentation from data collection to dataset construction. During this process, frames were extracted from each patient's video. Each frame was examined to display the expert's diagnosis on a standard plane. The labelled frames were converted to 224 × 224 binary images with zero scaling and padding. Subsequently, data augmentation was performed using only rotations. This is because, owing to the nature of the device, the magnification is fixed, and the rotation is not. Two rotation layers were added with up-down, left-right inversion, and random values between 0 and 360, increasing the total amount of data by six times. This process was applied only to the training set.

Figure 4(b) shows the AI classification method for the standard plane. There are many classification methods; however, a relatively lightweight CNN-based model has been adopted owing to resource limitations. A binary image of 224x224 was used as the input, as shown in Figure 4(a), and the binary classification results were obtained using sigmoid. The model was analyzed and compared with MobileNetV2, MobileNetV1, DenseNet121, EfficientNetV2B0, NASNetMobile, and ResNet50. Figure 4(c) is a schematic of AI model for hip landmark segmentation for the Graf algorithm. As in Figure 4(b), semantic segmentation was performed using the UNet-based model, which does not require a large amount of computation owing to resource limitations. A 224 × 224 binary image was used as the input and classified into four classes and output. The four classes are classified into Labrum, Bonny rim, Illium, and Lower rim. Therefore, a 4 channel area mask of 224 × 224 was output. The model obtains the results with NestedUNet, Unet, DeepLabV3Plus, and PAN and performs a dice coefficient comparison.

Figure 4(d) describes how the Graf algorithm is applied to the Graf hip area acquired in Figure 4(c). First, the center point was extracted from the labrum. Then, the baseline vector is extracted from the Illium, and as a method, the vector is extracted using the PCA algorithm. The orthogonal vector was then obtained and moved to the bony rim and lower limb to find the bony roof pointer at the end. The pointer extraction method for the bony and inferior rims was the same, and the baseline vector was moved vertically to find the end pointer. The Graf algorithm[13,14] was applied using four pointers and baseline vectors.

References

1. Fan, Z.-Z. *et al.* Risk factors for misdiagnosis in children with developmental dysplasia of the hip: a retrospective single centre study. *BMJ Paediatr. Open* **7** (2023).

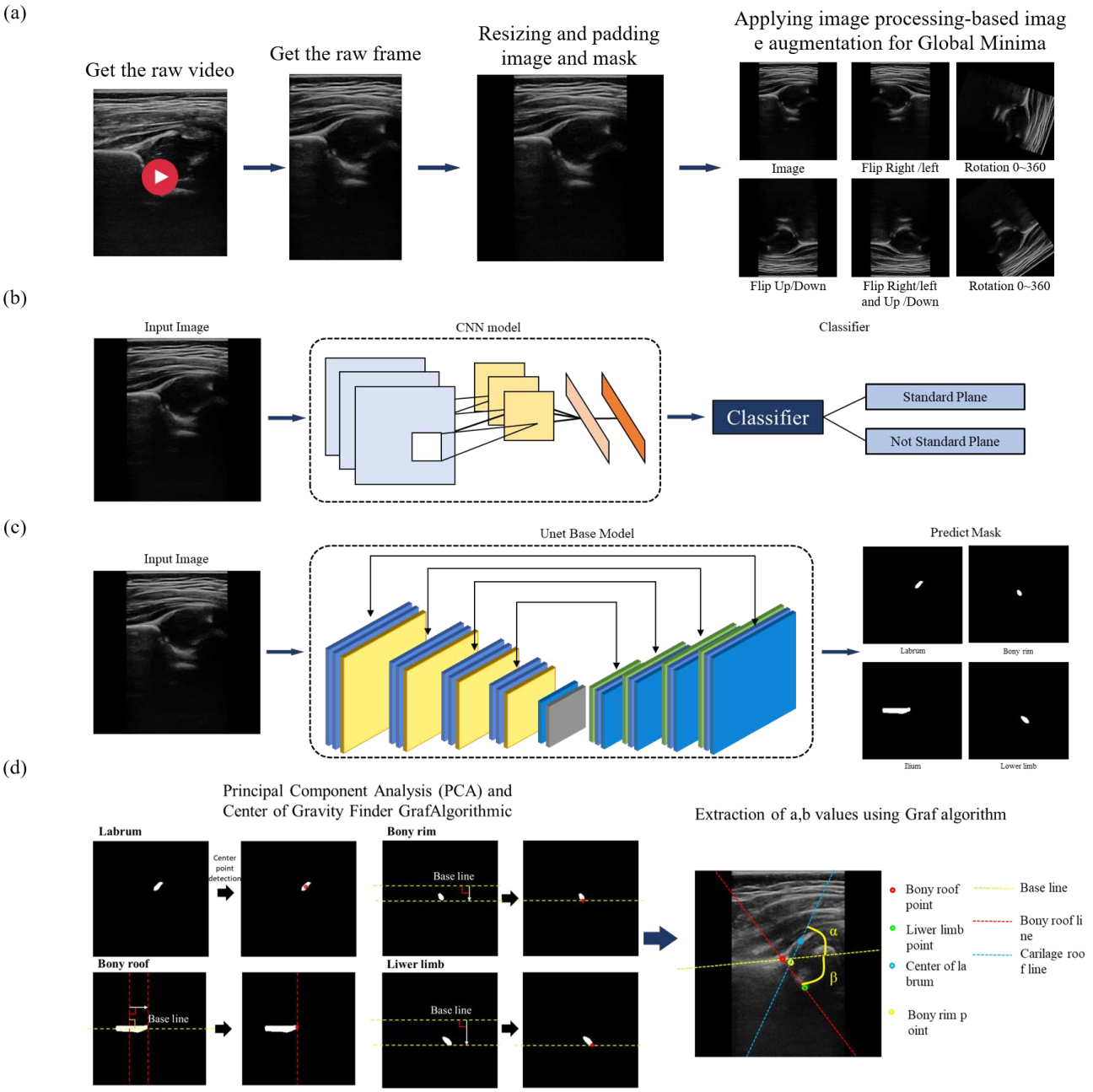


Figure 4. Overview of the entire process: (a) Image pre-processing and augmentation, (b) AI classification method for the standard plane, (c) Schematic diagram of landmark segmentation for the Graf algorithm, and (d) Method of applying the Graf algorithm in the Graf area segmentation mask.

2. Shaw, B. A. *et al.* Evaluation and referral for developmental dysplasia of the hip in infants. *Pediatrics* **138** (2016).
3. Ziegler, C. M. *et al.* Clinical examination and patients' history are not suitable for neonatal hip screening. *J. Child. Orthop.* **16**, 19–26 (2022).
4. Paton, R. W. Screening in developmental dysplasia of the hip (ddh). *surgeon* **15**, 290–296 (2017).
5. Dezateux, C. & Rosendahl, K. Developmental dysplasia of the hip. *The Lancet* **369**, 1541–1552 (2007).
6. Studer, K. *et al.* Increase in late diagnosed developmental dysplasia of the hip in south australia: risk factors, proposed solutions. *Med. J. Aust.* **204**, 240–240 (2016).

7. St George, J., Kulkarni, V., Bellemore, M., Little, D. G. & Birke, O. Importance of early diagnosis for developmental dysplasia of the hip: a 5-year radiological outcome study comparing the effect of early and late diagnosis. *J. Paediatr. Child Heal.* **57**, 41–45 (2021).
8. Broadhurst, C. *et al.* What is the incidence of late detection of developmental dysplasia of the hip in england?: a 26-year national study of children diagnosed after the age of one. *The bone & joint journal* **101**, 281–287 (2019).
9. Azzopardi, T., Van Essen, P., Cundy, P. J., Tucker, G. & Chan, A. Late diagnosis of developmental dysplasia of the hip: an analysis of risk factors. *J. Pediatr. Orthop. B* **20**, 1–7 (2011).
10. Luhmann, S. J., Bassett, G. S., Gordon, J. E., Schootman, M. & Schoenecker, P. L. Reduction of a dislocation of the hip due to developmental dysplasia: implications for the need for future surgery. *JBJS* **85**, 239–243 (2003).
11. Grill, F. *et al.* The pavlik harness in the treatment of congenital dislocating hip: report on a multicenter study of the european paediatric orthopaedic society. - (1988).
12. Kuitunen, I. *et al.* Incidence of neonatal developmental dysplasia of the hip and late detection rates based on screening strategy: a systematic review and meta-analysis. *JAMA Netw. Open* **5**, e2227638–e2227638 (2022).
13. Graf, R. Fundamentals of sonographic diagnosis of infant hip dysplasia. *J. Pediatr. Orthop.* **4**, 735–740 (1984).
14. Graf, R. Identifying the anatomy and checking the landmarks and tilting effects. *Hip Sonogr. Diagn. Manag. Infant Hip Dysplasia* 27–30 (2006).
15. Õmeroğlu, H. Use of ultrasonography in developmental dysplasia of the hip. *J. children's orthopaedics* **8**, 105–113 (2014).
16. Tschauner, C. & Matthissen, H. Hip sonography with graf-method in newborns: Checklists help to avoid mistakes. *OUB* **1**, 7–8 (2012).
17. Jaremko, J. L. *et al.* Potential for change in us diagnosis of hip dysplasia solely caused by changes in probe orientation: patterns of alpha-angle variation revealed by using three-dimensional us. *Radiology* **273**, 870–878 (2014).
18. Alam, M. E., Smith, N., Watson, D., Hassan, T. & Neupane, K. Early screening of ddh using svm classification. In *2019 SoutheastCon*, 1–4 (IEEE, 2019).
19. Jaremko, J. L. *et al.* Ai aided workflow for hip dysplasia screening using ultrasound in primary care clinics. *Sci. Reports* **13**, 9224 (2023).
20. Ronneberger, O., Fischer, P. & Brox, T. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18*, 234–241 (Springer, 2015).
21. Altman, D. G. Why we need confidence intervals. *World journal surgery* **29**, 554–556 (2005).
22. Guindon, B. & Zhang, Y. Application of the dice coefficient to accuracy assessment of object-based image classification. *Can. J. Remote. Sens.* **43**, 48–61 (2017).

Acknowledgements (not compulsory)

This research was funded by the Korea Medical Device Development Fund grant funded by the Korean government (Ministry of Science and ICT, Ministry of Trade, Industry and Energy, Ministry of Health & Welfare, Ministry of Food and Drug Safety) (Project Number: 202011B06-02).

Author contributions statement

Young-seop Lee conceived the experiment, conducted the experiment, and analyzed the results. Gwang-gi Kim is the corresponding author, and Jeong Won Ryu,Young-jae Kim checked the results and provided guidance. All authors reviewed the manuscript.

Additional information

Correspondence and requests for materials should be addressed to Kwang Gi Kim .

data availability statement

Sequence data supporting the findings of this study are available with permission from the corresponding author